MIL-T-18912A(OS)
14 October 1976
SUPERSEDING
MIL-T-18912(NOrd)
29 July 1955

MILITARY SPECIFICATION

TUBING, STEEL, FOR 3.25-INCH AND 5.0-INCH O.D. ROCKET MOTORS

This specification is approved for use by The Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cold-drawn seamless steel tubing for use on 3.25-inch and 5.0-inch O.D. rocket motors.

1.2 Classification. Tubing shall be of one type only and of the following sizes, as specified (see 6.2).

Size - 5.0-inch O.D.
Size - 3.25-inch O.D.

* 2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issues in effect on date of invitation for bids or request for proprosals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

SPECIFICATIONS

Military

MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Solid Application

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. The tubing furnished under this specification shall be a product which has been tested and has passed the preproduction tests specified herein.

* 3.1.1 Conditions for new preproduction tests. Any significant change in composition of material or in process of manufacture, which, in the opinion of the procuring activity, may affect high-temperature characteristics, shall necessitate new preproduction tests in accordance with 4.3.1.

3.2 Material. Tubing covered by this specification shall be cold-drawn, seamless steel tubing which has been given such treatment as may be necessary to attain the required mechanical properties.

3.2.1 Mechanical properties at 900° F. The tubing shall have an ultimate strength of 75,000 pounds per square inch (psi) minimum at 900° F, when determined as specified in 4.3.1 and 4.4.1.

3.2.2 Mechanical properties at ambient temperature. The tubing shall have an ultimate strength of 100,000 psi minimum at ambient temperature. Yield strength (0.01 percent offset) shall be 80,000 psi minimum. The elongation in 2 inches shall be 13 percent minimum when determined as specified in 4.4.2.

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* 3.2.3 Chemical composition. The chemical composition of the steel used shall be submitted to the procuring activity for record. The point range of each element, for the steel submitted, shall be not greater than that for a corresponding SAE or AISI steel. All alloying elements present in significant quantity, as well as carbon, manganese, sulphur, phosphorus, and silicon, shall be included.

3.3 Form. Dimensions, tolerances, concentricity, straightness, out of roundness, and other details of form shall be such as to permit manufacture in accordance with the drawings for the finished rocket motors or as defined by the contract.

3.4 Product marking. Each piece shall be marked as required by the contract or order.

3.5 Workmanship. Tubing shall have a good workmanlike finish conforming to the best practice required for comparable quality tubing. Tubing shall be smooth, clean, and free from heavy scale or oxide on the interior and exterior surfaces, and shall be free from burrs, seams, tears, grooves, laminations, slivers, pits, and other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection lot. As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in inspection lots shall be as determined by the procuring activity and may differ from the quantity designated in the contract or order as a lot for production, shipment, of other purpose.

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4.1.2 Lot for tensile test. For purposes of tensile testing, a lot shall consist of all tubes of the same size, heat, and quality which have received the same heat treatment or stress relief treatment under the following conditions:

4.1.2.1 For a batch-type furnace. All tubes in one furnace charge.

4.1.2.2 For a continuous-type furnace. All tubes in one uninterrupted interval not exceeding 8 hours duration, up to a maximum quantity of 20 tons.

4.1.3 Lot for chemical analysis. For purposes of chemical analysis, a lot shall consist of a single heat of steel.

4.2 Sampling.

* 4.2.1 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures shall be in accordance with the provisions of MIL-STD-105.

* 4.2.2 Sampling for preproduction tests. Two 24-inch lengths of tubing, selected from different tubes of the initial manufacture under each contract, shall be submitted at the contractor's expense to the laboratory designated in the contract or purchase order for testing as specified in 4.3.1.

4.2.3 Sampling for mechanical properties at 900° F. From every 100 tons, minimum, of tubing manufactured in accordance with the contract, two 24-inch lengths shall be selected for check testing as specified in 4.4.1.

4.2.4 Sampling for tensile tests at ambient temperature. Specimens for tests at ambient temperature as specified in 4.4.2 shall be taken from two tubes from each lot, see 4.1.2.

4.2.5 Sampling for chemical analysis. Two samples shall be taken for chemical analysis from each heat. Chips for a chemical sample may be taken from several sections around a tube, or from broken tensile test specimens.

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4.3 Preproduction inspection. Preproduction inspection shall consist of the following:

4.3.1 Strength at 900° F. Initial manufacture of tubing under each contract shall be tested at government expense, to determine compliance with 3.2.1. The rate for tensile tests at 900° F shall be one-half inch per minute free-running (no load) cross-head speed.

* 4.3.2 Submission of data. Manufacturing data of the tubing being submitted for preproduction tests shall include mechanical properties at ambient temperature, chemical analysis, manufacturing process, and heat treatment. These data shall be submitted to the procuring activity for transmittal to the testing laboratory as designated in the contract or purchase order.

4.4 Quality conformance inspection.

4.4.1 Check test. After satisfactory preproduction tests, check testing by the same procedure as specified in 4.3.1 shall be conducted on samples selected in accordance with 4.2.3, to determine compliance with 3.2.1.

* 4.4.2 Strength at ambient temperature. Test specimens taken in accordance with 4.2.4 shall be machined to form the dimensions specified by the procuring activity. From each tube selected for test, two specimens shall be removed at positions 1800 apart within the same longitudinal tube location. The cylindrical surfaces of the central part of the test specimen which includes the gage portion and extends 1 inch past the intersections of the fillets with the grip portions at each end, shall be left undisturbed (i.e., not machined, bent, flattened, or ground). The cross sectional area of the specimen shall be calculated by multiplying the minimum wall thickness by the least width of the gage section as measured by a micrometer. The specimens shall be tested to determine compliance with 3.2.2.

4.4.3 Chemical analysis. A chemical analysis of drillings obtained as specified in 4.2.5 shall be made to determine compliance with 3.2.3.

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* 5. PACKAGING

5.1 Preservation-packaging. When required by the contract or order, tubing shall be coated inside and out with rust preventive compound as specified in MIL-C-16173, grade 3.

5.2 Packing. Tubing shall be loose or in lifts as required by the contract or order.

5.3 Marking. In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Tubing supplied under this specification is intended for use on rocket motors.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Size required (see 1.2)
c. Product marking (see 3.4)
d. Applicable drawings (see 3.3)
e. Laboratory to perform preproduction tests (see 4.2.2)
f. Submission of data (see 4.3.2)
g. Test specimen dimensions (see 4.4.2)
h. Preservation, if required (see 5.1)
i. Packing, loose or in lifts (see 5.2).

* 6.3 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:

NAVY - OS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE MIL-T-18912A(OS) - Tubing, Steel, For 3.25-Inch and 5.0-Inch O.D. Rocket Motors

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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